

Solutions & Marking Scheme: Coordination Chemistry

Official Step-by-Step Marking Scheme by Ajay Choudhary Sir

Chapter / Focus Area: Inorganic Chemistry Solutions

Marking Scheme & Model Answers

Ans 1. Pentaamminecarbonatocobalt(III) chloride and Diamminechloridonitrito-N-platinum(II). [2 Marks]

Ans 2. $[\text{Ni}(\text{CN})_4]^{2-}$: CN^- is strong field ligand, pairs 3d electrons $\rightarrow$ dsp^2 (Square planar, Diamagnetic).

$[\text{NiCl}_4]^{2-}$: Cl^- is weak field ligand, no pairing $\rightarrow$ sp^3 (Tetrahedral, Paramagnetic with 2 unpaired electrons). [4 Marks]

Ans 3. Weak field ($\Delta_o < P$): $t_{2g}^3 e_g^1$. Strong field ($\Delta_o > P$): $t_{2g}^4 e_g^0$. [3 Marks]

Ans 5. Ti^{3+} has 3d1 configuration (single d-electron undergoes visible light excitation from t_{2g} to e_g). Sc^{3+} has 3d0 configuration with no d-electrons, hence colorless. [3 Marks]

Official Verified Study Resource • Mentored by Ajay Choudhary Sir

Web: <https://pankaj2816.github.io/ajay-choudhary-chemistry/> • Contact: contact@ajaychemistry.com